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Research article

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Nuclear transfer of a beneficial fungal effector-host target complex via the NTF2 pathway underpins symbiont-induced plant immunity

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The author responsible for distribution of materials integral to the findings presented in this article in accordance with the policy described in the Instructions for Authors (https://academic.oup.com/plphys/pages/General-Instructions) is: Weixing Shan.

Abstract

The symbiotic fungus *Serendipita indica* confers broad-spectrum beneficial effects on diverse plant hosts. Its key effector SIE141 elicits immunity against *Phytophthora* and salt tolerance by binding and relocalizing thioredoxin CDSP32 from chloroplast to the nucleus. Here, we show that this functionally essential nuclear transfer process of the SIE141-CDSP32 complex is mediated by the host NTF2 proteins. NTF2 family proteins are direct targets of SIE141, whose knockdown abolished nuclear accumulation of both SIE141 and CDSP32, leading to their rendered accumulation to chloroplasts. The glutamine residue at position 40 of NbNTF2A is critical for its interaction with both SIE141 and Ran-GTPase 1. SIE141 modulates NbNTF2-NbRan1 interaction in a dose-dependent manner, without impairing nuclear accumulation of NbRan1 and the NTF2-mediated positive immune function of Ran1. Our results reveal a previously unknown mechanism where a beneficial symbiotic fungal effector utilizes the conserved NTF2 rather than other core nuclear import machinery components to traffic an immune complex and enhance disease resistance. These findings provide a potential strategy for engineering plant immunity by manipulating NTF2–Ran-GTPase dependent nucleocytoplasmic transport.

Introduction

Plant beneficial symbionts establish highly diverse and dynamic interactions with plants and—similar to the nutritional health effects of animal gut flora—improve plant nutrient acquisition, growth, disease resistance, and abiotic stress tolerance (Trivedi et al. 2020). The success of beneficial microbes in colonizing plants depends on close communication and manipulation of host plants (Rodriguez et al. 2019). Intriguingly, like pathogens, beneficial microbes rely on effector proteins for this purpose (Stone et al. 2022).

Serendipita indica, a beneficial endophyte exerting a diversity of beneficial effects in broad range of host plants, releases effectors (defined as *S. indica* effectors [SIEs]) in the process of host colonization and activation of beneficial plant effects. Studying these effectors therefore offers a chance to understand the molecular mechanisms involved in the establishment of symbioses with a broad spectrum of hosts and in the activation of beneficial effects (Rafiqi et al. 2013). Some SIEs were described as suppressors for plant immunity, eg an SIE targeting FGB1 deregulates innate sensing of β-glucans in plants and effector Dld1 interferes with oxidative stress (Wawra et al. 2016; Nostadt et al. 2020). Conversely, certain effectors cause unexpected immunostimulatory effects. SIE141 not only primes defense responses against *Phytophthora* pathogens, as causal disease agents in multiple crops, but also enhances abiotic stress tolerance. Nuclear localization seemed to be essential for this dual functionality (Zhang et al. 2024). Previous research also revealed that SIE141 targets

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thioredoxin-like CDSP32 and re-localizes this protein to the nucleus, where it could also contribute to the activation of local and systemic plant immunity (Zhang et al. 2024).

The nuclear translocation of immune-related proteins, mediated by importins and nucleoporins, facilitates transcriptional reprogramming through coordinated action of transcriptional regulators or directed transport of nuclear-localized Resistance (R) proteins, to orchestrate plant defense signaling, highlighting the central role of nuclear trafficking in plant immunity (Liu and Coaker 2008; Motion et al. 2015). Thus, effectors target and highjack components of this trafficking system to promote infection. For instance, MoHTR1, a nuclear effector of the rice blast fungus *Magnaporthe oryzae* is translocated to rice nuclei via binding rice importin α (Lim et al. 2024).

The nuclear transport factor 2 (NTF2) was identified to work together with Ras-family GTPase (Ran) to mediate nuclear trafficking of macromolecules (Clarkson et al. 1997). NTF2 undergoes dimerization and binds Ran-GDP to facilitate its nuclear import, where Ran converts to its GTP-bound state via guanine nucleotide exchange factors (GEFs), generating a gradient that drives export processes (Chaillan-Huntington et al. 2001). Following GTP hydrolysis, cytoplasmic Ran-GDP is recycled by NTF2, ensuring sustained nuclear-cytoplasmic transport (Clarkson et al. 1997; Ribbeck et al. 1998). NTF2 exhibits multifaceted roles ranging from immune regulation to stress adaptation across diverse biological systems. In mammals, NTF2 dysregulation disrupts core cellular processes in tumor microenvironments, including cell cycle control, DNA repair mechanisms, and nucleocytoplasmic transport (Zhang and Gao 2021; Li et al. 2022). Emerging evidence further implicates a function of NTF2 in host–microbe interactions. The bacterial *Mycobacterium tuberculosis* effector PtpA directly targets Ran and exploits the Ran-GDP–NTF2 complex, where NTF2 enhances the PtpA-Ran interaction to facilitate nuclear entry, ultimately inducing host cell ferroptosis and enhancing virulence (Qiang et al. 2023). In turn, different plant systems reveal NTF2-associated defense strategies. Wheat TaNTF2 activation during *Puccinia striiformis* infection enhances biotrophic pathogen resistance (Zhang et al. 2018). Similarly, NTF2 domain-containing *GmPRP* was upregulated in soybean leaves in response to *Phytophthora sojae* infection, and its recombinant protein displayed antimicrobial activity and significantly inhibited hyphal growth of *P. sojae* in vitro (Jiang et al. 2015). NTF2 was also reported to coordinate abiotic stress adaptation. Alfalfa (*Medicago sativa*) MsNTF2L was reported to enhance plant drought tolerance via integrated regulation of stomatal dynamics, reactive oxygen species scavenging, and photosynthetic efficiency (Luo et al. 2022). These findings collectively indicate that NTF2 family proteins serve as key nodes in host–microbe interactions and as a central hub interfacing cellular homeostasis with environmental perception.

In this study, we report that plant disease resistance induced by the symbiotic fungal effector SIE141, from the endophyte *Serendipita indica*, requires host NTF2 family proteins. NbNTF2 proteins facilitate the nuclear transfer of SIE141 and its target CDSP32 for SIE141-mediated activation of plant immunity. Furthermore, NbNTF2 family proteins exhibit divergent immune functions, with NbNTF2A identified to be a negative immune regulator whose 40th glutamine residue at the N-terminus is required for both its interaction with SIE141 and its immune function. This site also mediates its binding to NbRan1, thereby enabling its nuclear translocation and supporting its immunity function against *P. parasitica*. SIE141 modulates NbNTF2-NbRan1 interactions in a dose-dependent manner, without impairing the nuclear accumulation of NbRan1, thus retaining its NTF2-mediated immunity-activating function. These findings refine models of symbiotic effector-mediated immunity, especially how beneficial fungal effectors recruit a central plant nuclear-cytoplasm trafficking system and inform engineering strategies for enhancing crop disease resistance.

Results

SIE141 targets host plant NUCLEAR TRANSPORT FACTOR 2 (NTF2) family proteins

Our previous study revealed that SIE141 stimulated plant immunity against *Phytophthora* pathogens and enhanced salt tolerance. Its nuclear localization was found to be required for its immune function (Zhang et al. 2024). To further understand mechanisms underlying SIE141-mediated plant immune activation, we performed immunoprecipitation–mass spectrometry (IP-MS)-based screening for its interacting host plant proteins in *Nicotiana benthamiana*. This led to the identification of multiple candidate target proteins, including NUCLEAR TRANSPORT FACTOR 2 (NbNTF2) family proteins (Table S1). Three family members (NbNTF2A, NbNTF2B, and NbNTF2C) were identified in *N. benthamiana* and cloned for further tests (Fig. 1a). The NTF2 family proteins are highly conserved across different kingdoms (Corbett and Silver 1996). Structural analyses using AlphaFold 3 showed that NbNTF2 proteins share similar protein structures with yeast (*Saccharomyces cerevisiae*) NTF2 (ScNTF2) and rat (*Rattus norvegicus*) NTF2 (RnNTF2) (Figure S1a), suggesting their conserved function. NbNTF2C lacks a 40-amino-acid segment at its C-terminus as compared to NbNTF2A and NbNTF2B, corresponding to 2 predicted terminal β-sheets (βe and βf) (Figure S1b). Subcellular localization analyses revealed that NbNTF2A and NbNTF2C localized exclusively to the nucleus, whereas NbNTF2B exhibited both nuclear and chloroplast localization (Figures S1c–S1f).

Yeast two-hybrid (Y2H), split-luciferase, bimolecular fluorescence complementation (BiFC), and co-immunoprecipitation (co-IP) assays demonstrated that SIE141 targets NbNTF2A and NbNTF2B but not NbNTF2C proteins (Fig. 1b–1e, Figure S1g–S1i). The integrity and expression of the NbNTF2 and SIE141 fusion proteins were confirmed by Western blot analysis (Figure S1g).

NbNTF2 proteins mediate nuclear transfer of SIE141

In order to find out whether NbNTF2 proteins participate in SIE141 nuclear import, the conserved coding sequence was selected for generating TRV-*NbNTF2* plants. Reverse transcription quantitative real-time PCR (RT-qPCR) was utilized to confirm whether all 3 *NbNTF2* family genes were knocked down (Figure S2a). Subsequently, confocal microscopy was conducted with *NbNTF2*-silenced or *GUS*-silenced *N. benthamiana* leaves expressing *GFP*-fused *SIE141*. The GFP signal of SIE141, but not histone-specific mCherry signal, was significantly reduced in the *NbNTF2*-silenced as compared to *GUS*-silenced *N. benthamiana* leaves (Fig. 2a and 2b). Meanwhile, free GFP was also reduced upon *NbNTF2* silencing as compared to *GUS* silencing (Fig. 2a and 2b), suggesting that NTF2 proteins also mediate free GFP transfer to the nucleus. In addition, proteins from nuclear fraction were extracted and detected for the level of free GFP or SIE141 in nuclei of different samples by SDS-PAGE assay. These analyses showed that both free GFP and SIE141-GFP were reduced in *NbNTF2*-silenced leaves as compared to *GUS*-silenced leaves (Fig. 2c), while no difference in Flag-GFP (FGFP) or SIE141-GFP protein abundances were evident in the cytoplasmic fraction (Fig. 2c). These

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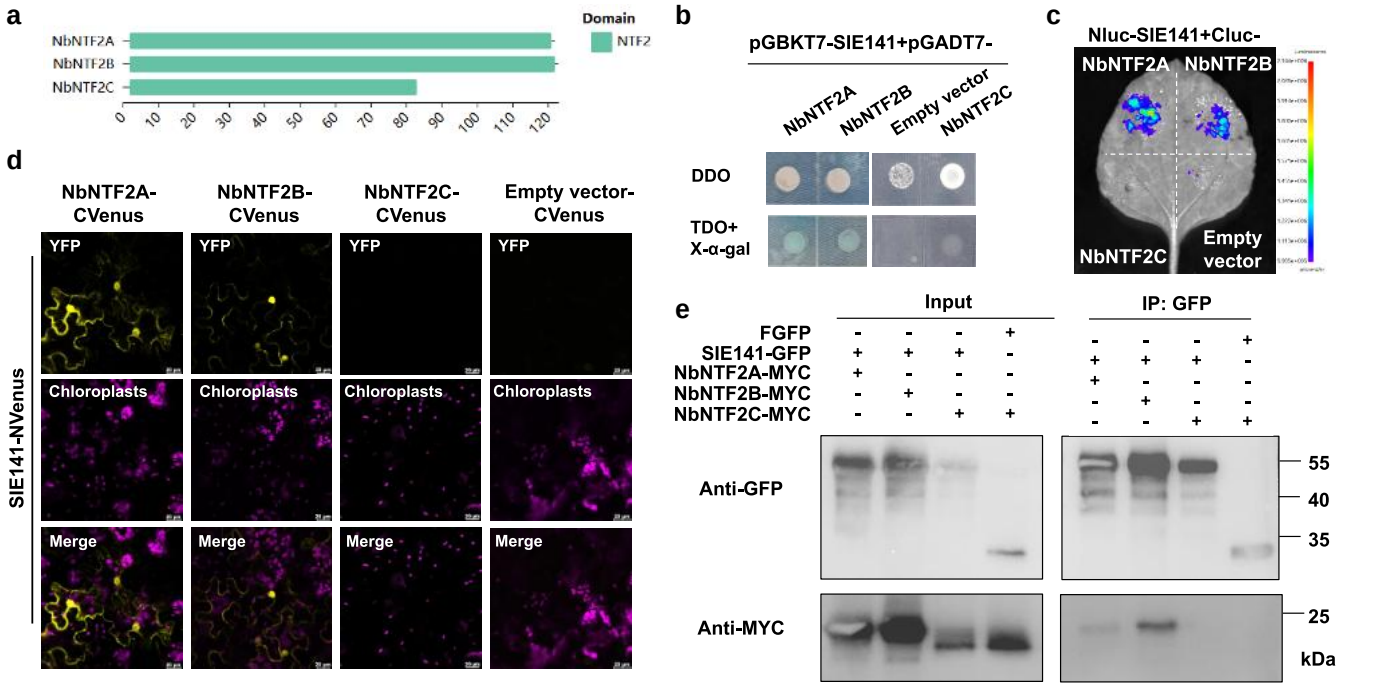


Figure 1 SIE141 targets NbNTF2A and NbNTF2B but not NbNTF2C. (a) Schematic graph of NbNTF2 family members showing their conserved domains according to NCBI Batch CDD-Search. Green boxes indicate the nuclear transport factor 2 (NTF2) domain. The graph was created by Chiplot. (b) The yeast strain AH109 co-expressing SIE141 with NbNTF2A or NbNTF2B on SD/-Leu-Trp-His medium exhibited α-galactosidase activity while co-expressing SIE141 with empty vector or NbNTF2C did not show α-galactosidase activity. (c) Luciferase complementation assay to detect interactions between SIE141 and NbNTF2A, NbNTF2B, or NbNTF2C. Nluc-SIE141 were co-expressed with Cluc-NbNTF2A, Cluc-NbNTF2B, or Cluc-NbNTF2C via agroinfiltration. Nluc-SIE141 + Cluc-NbNTF2A or Nluc-SIE141 + Cluc-NbNTF2B yield luciferase activity which was not detected with Nluc-SIE141 + Cluc-NbNTF2C. (d) Bimolecular fluorescence complementation (BiFC) assay to detect interactions between SIE141 and NbNTF2A, NbNTF2B, or NbNTF2C in *N. benthamiana*. CVenus-tagged NbNTF2A, NbNTF2B, or NbNTF2C were co-expressed with NVenus-tagged SIE141 via agroinfiltration. Bars=20 μm. (e) Co-immunoprecipitation (co-IP) assays with GFP-Trap for GFP-fused SIE141, MYC-fused NbNTF2 family members (NbNTF2A, NbNTF2B, and NbNTF2C) were detected by the MYC-specific antibody. Flag-GFP (FGFP) was used as the negative control. Protein markers indicate size in kDa.

results indicated that the reduction of both SIE141-GFP signal intensity and protein abundance in the nucleus was caused by the absence of NbNTF2 but not a general reduction of protein expression.

On the other hand, when *SIE141-GFP* was overexpressed together with *NbNTF2A-mCherry*, the GFP fluorescence of SIE141 was also significantly decreased in nuclei as compared to co-expressed with mCherry (Figures S2b and S2c). The reduced abundance of GFP in the separated nuclear fractions (Figure S2d) further suggested that certain expression levels of NbNTF2A are important for nuclear trafficking of SIE141.

NbNTF2 is required for SIE141-induced nuclear translocation of NbCDSP32

Our previous study revealed that SIE141 binds the chloroplast transit peptide (CTP) of NbCDSP32 and blocks its chloroplast-targeting function, thereby inducing relocation of NbCDSP32 from chloroplasts to nuclei (Zhang et al. 2024). To investigate whether this altered nuclear localization of NbCDSP32 depends on NTF2 family proteins, we first examined its nuclear-targeted CTP-truncated NbCDSP32 mutant (ctpNbCDSP32-GFP). *GFP*-fused *NbCDSP32* and *ctpNbCDSP32* constructs were respectively co-expressed with *Histone-mCherry* in *GUS*-silenced or *NbNTF2*-silenced *N. benthamiana* leaves. Confocal microscopy showed nuclear localization of GFP-tagged ctpNbCDSP32 in *GUS*-silenced leaves but not in *NbNTF2*-silenced samples (Figures S3a and S3b). This localization pattern was confirmed in immunoblot analyses using nuclear and cytoplasmic fractions (Figure S3c) indicating that nuclear transfer of ctpNbCDSP32 required NbNTF2.

Next, nuclear re-localization of the full-length NbCDSP32 triggered by SIE141 was examined for its dependency on NbNTF2. In TRV-*GUS* control plants, co-expression of *SIE141-mCherry* induced nuclear accumulation of NbCDSP32-GFP, in contrast to the mCherry control (Fig. 3a). However, in TRV-*NbNTF2* plants, NbCDSP32-GFP remained exclusively in chloroplasts, even in the presence of SIE141-mCherry (Fig. 3b). Subcellular fractionation followed by immunoblotting confirmed that NbCDSP32 was undetectable in nuclear fractions from *NbNTF2*-silenced leaves, particularly when *SIE141* was co-expressed (Fig. 3c). These results are consistent with the absence of nuclear ctpNbCDSP32-GFP in TRV-*NbNTF2* plants further demonstrating that NbNTF2 is indispensable for SIE141-mediated nuclear translocation of NbCDSP32.

Intriguingly, when nuclear import was blocked by *NbNTF2*-silencing, SIE141-mCherry itself lost its normal nucleocytoplasmic distribution and was redirected to chloroplasts upon co-expression with *NbCDSP32-GFP*; in contrast to *Flag-GFP* co-expressed in *NbNTF2*-silenced plants (Fig. 3b). It suggests that when nuclear access is blocked, SIE141 and its target NbCDSP32 are retained together in chloroplasts, where NbCDSP32 normally resides.

Extending this regulatory interplay, the nuclear localization of SIE141 is also influenced by the silencing of *NbCDSP32*. Little difference of GFP fluorescence could be detected in *NbCDSP32*-silenced plants as compared to *GUS*-silenced samples (Figures S4a–S4c).

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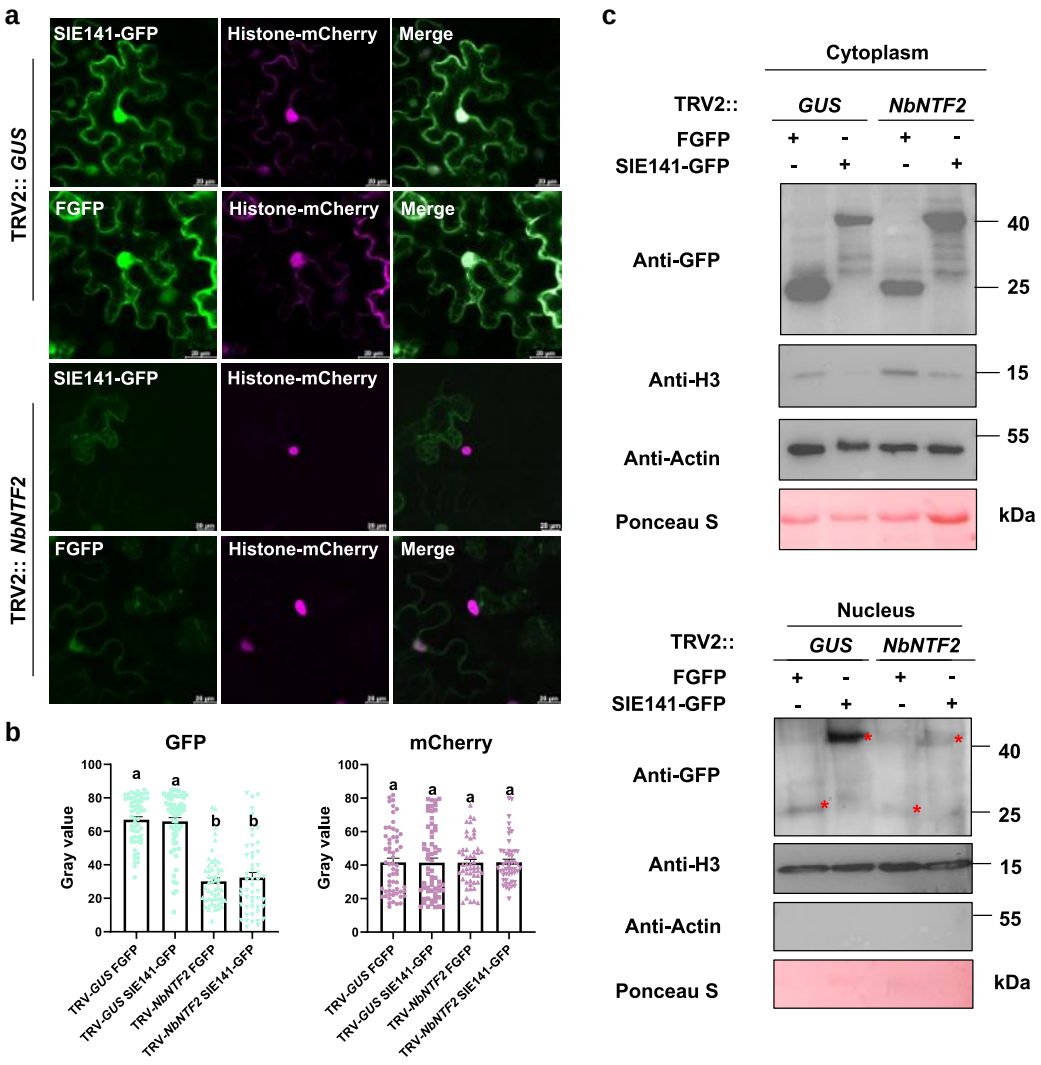


Figure 2 NbNTF2 is involved in the nuclear transport of SIE141. (a) Flag-GFP (FGFP) and SIE141-GFP were transiently expressed in *GUS*-silenced or *NbNTF2*-silenced *N. benthamiana* leaves via agroinfiltration. Confocal microscopy was conducted at 48 hpi. Bars=20 μm. (b) Fluorescence intensity of Flag-GFP and SIE141-GFP signals and Histone-mCherry in individual nuclei ($n \geq 50$) from *GUS*-silenced or *NbNTF2*-silenced samples. Data were analyzed with one-way ANOVA (Brown-Forsythe and Welch tests), different letters indicate significant difference at $P < 0.05$. (c) Protein level of SIE141-GFP was decreased in *NbNTF2*-silenced samples as compared to *GUS*-silenced samples. Immunoblot analysis for the detection of SIE141-GFP abundance in nuclear extracts from *GUS*-silenced or *NbNTF2*-silenced leaves using a GFP antibody. Histone 3 (H3) antibody was used as internal reference to confirm equal loading of nuclear proteins. Actin antibody and Ponceau S were used as external references.

SIE141-GFP protein was barely detectable in nuclear fractions of *NbCDSP32*-silenced samples, suggesting that nuclear transfer of SIE141 is partially dependent on its target NbCDSP32 (Figure S4d).

These results suggest that NbNTF2 family proteins are not only required for the nuclear localization of SIE141 but are also essentially involved in SIE141-induced nuclear re-localization of CDSP32.

A glutamine residue of NbNTF2A is critical for interactions with both SIE141 and NbRan GTPase

Transcriptome analyses revealed that both *P. infestans* and *P. parasitica* induced all 3 *NbNTF2* family members, suggesting their role in plant immunity (Figures S5a and S5b, Tables S3 and S4). To examine their potential immune function, MYC-fusion constructs were generated for *NbNTF2A*, *NbNTF2B*, and *NbNTF2C* and transiently expressed in *N. benthamiana* for subsequent pathogen infection assays. The expression and integration of fusion proteins were determined by Western blotting (Figure S5c). *Flag-GFP* (*FGFP*) or individual *NTF2* members were transiently expressed using agroinfiltration assays in *N. benthamiana* leaves, followed by inoculation with *P. parasitica* zoospores. Pathogen colonization was quantified at 48 hours postinoculation (hpi). *NbNTF2A* expression promoted pathogen colonization as compared to *FGFP*, whereas *NbNTF2C* expression conferred enhanced resistance, as evidenced by reduced pathogen biomass and lesion diameters (Fig. 4a–4c). In addition, *NbNTF2B* expression displayed no significant immune phenotype (Fig. 4a–4c).

Despite selecting different fragments from the untranslated (UTRs) or coding sequence regions (CDS) to perform virus-induced gene silencing (VIGS) assays, the high level of sequence conservation within the NbNTF2 family members made it impossible to specifically silence

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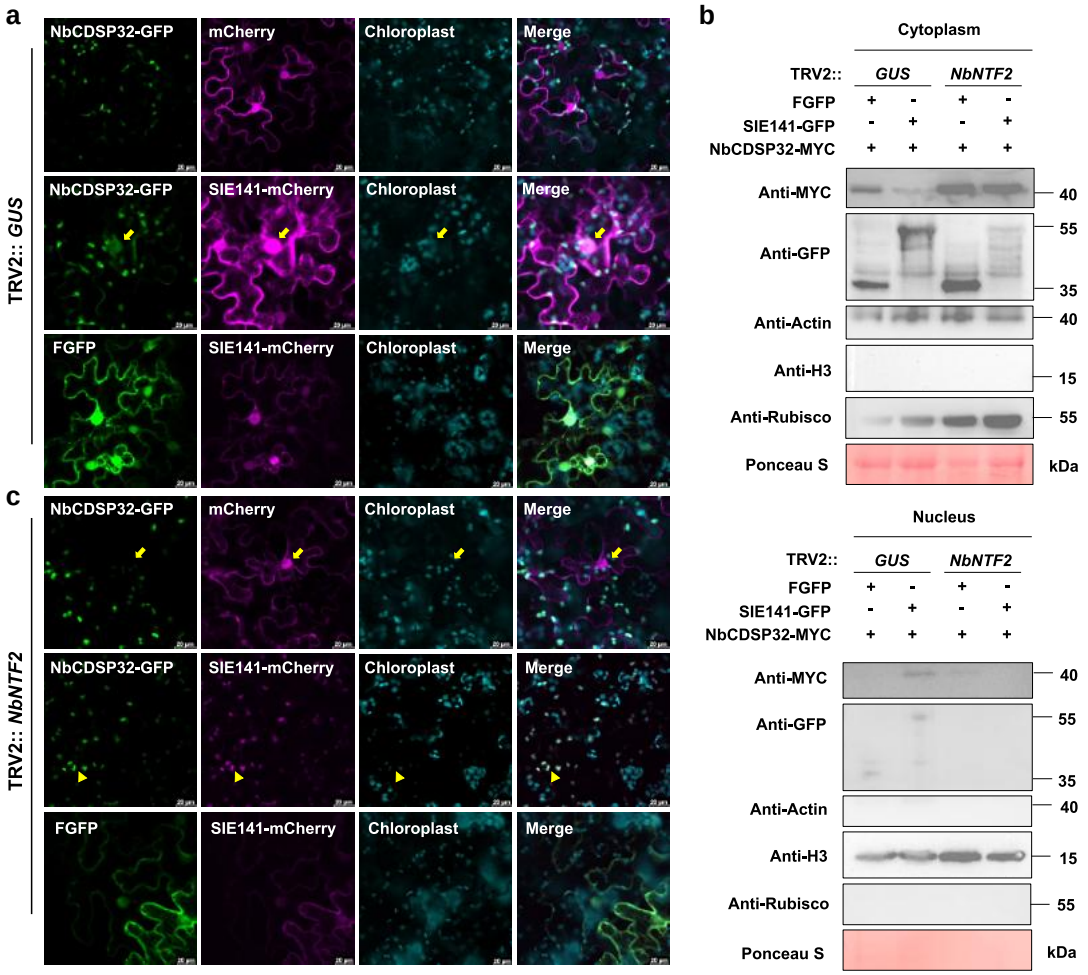


Figure 3 NbNTF2 is required for SIE141-induced nuclear translocation of NbCDSP32. (a-b) NbCDSP32-GFP was transiently co-expressed with SIE141-mCherry or mCherry via agroinfiltration in (a) *GUS*-silenced or (b) *NbNTF2*-silenced *N. benthamiana* leaves. Flag-GFP (FGFP) was co-expressed with SIE141-mCherry as additional negative control. Confocal microscopy was conducted at 48 hpi. Yellow arrows indicate nuclear localization of NbCDSP32 when co-expressed with SIE141 in *GUS*-silenced leaves. Yellow triangles indicate chloroplast localization of SIE141 when co-expressed with NbCDSP32 in *NbNTF2*-silenced leaves. Bars=20 μm. (c) Immunoblot analysis for the detection of NbCDSP32-4MYC abundance in nuclear and cytoplasmic extracts in *GUS*-silenced or *NbNTF2*-silenced leaves co-expressing SIE141-GFP or FGFP. MYC antibody detected NbCDSP32-4MYC and GFP antibody detected SIE141-GFP or FGFP protein levels in different fractions of individual samples. Ponceau S was used to verify the equal loading of proteins. Actin, Histone 3 (H3), and Rubisco antibodies were used as external references.

individual members (Figures S5d–S5f). All tested VIGS constructs resulted in co-silencing of *NbNTF2* family members (Figures S5d–S5f). Yet, at 48 hpi with *P. parasitica* zoospores, silenced plants showed no significant differences in pathogen biomass relative to *GUS*-silenced control plants, revealing no detectable change in resistance or susceptibility (Fig. 4d).

The interaction between NTF2 and Ran-GDP is important for nuclear protein trafficking (Clarkson et al. 1997). We therefore examined the relationship between immune function of NTF2A and the Ran-mediated nucleus-cytoplasm trafficking. The conserved glutamic acid (E) at position 40 in NTF2 was previously found to mediate this interaction in yeast (Clarkson et al. 1997; Zhang and Gao 2021), which corresponds to the glutamic acid at position 38 in the NbNTF2 family protein members. This residue was mutated to generate NbNTF2A^{E38K} (Figure S1b). However, AlphaFold 3 prediction showed that the interaction between NbRan1 and NbNTF2A^{E38K} retained, with an interface predicted template modeling (ipTM) score of 0.83; a high accuracy prediction value for protein-protein interactions (Figure S5g). Consistently, luciferase, BiFC and co-IP assays confirmed that the NbNTF2A^{E38K} mutant retained interaction with NbRan1, and with SIE141 (Fig. 4h–4m, Figures S5j–S5k). Transient overexpression assays in *N. benthamiana* showed that NbNTF2A^{E38K} retained its negative immune function, as indicated by rendering increased plant susceptibility and enhanced pathogen biomass as compared to the control (Figs 4e–4g).

Although mutation of the conserved E38 residue of NbNTF2A did not disrupt its interaction with NbRan1 nor its susceptibility-promoting function (Fig. 4), we sought to identify paralog-specific residues that might account for effector association and immune outcomes. Sequence alignments of the NTF2-like domain revealed a variable residue at position 40, where NbNTF2A uniquely possesses glutamine (Q40), NbNTF2B a valine (V40) and NbNTF2C a glutamic acid (E40) (Figure S1b). This paralog-specific Q40 in NbNTF2A was selected as a candidate for modulating both Ran/SIE141 interaction and immunity. Alanine or glutamic acid substitution mutants NbNTF2A^{Q40A} and NbNTF2A^{Q40E}, respectively, were generated to

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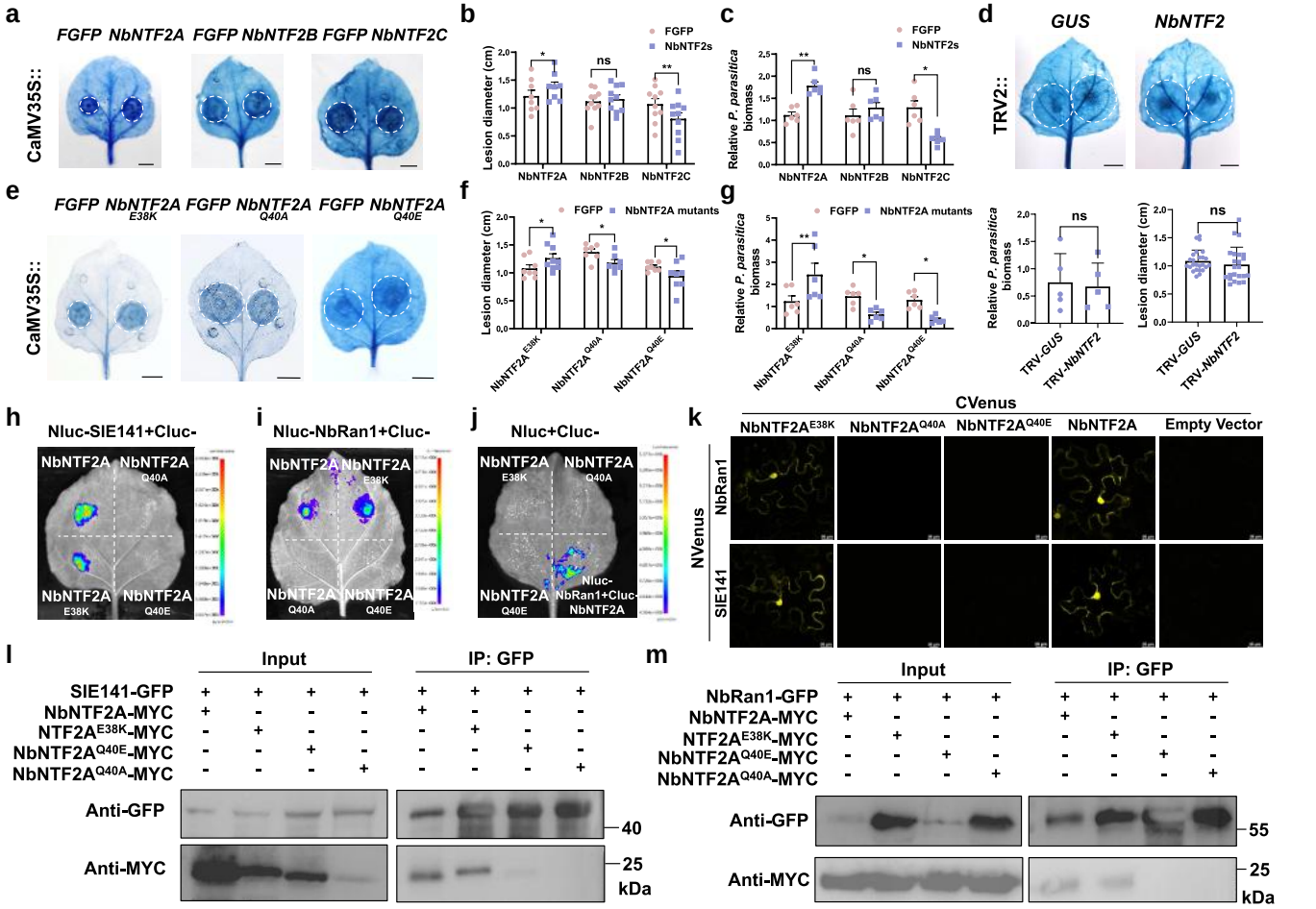


Figure 4 The 40th glutamine residue of NbNTF2A is critical for interactions with both SIE141 and Ran-GTPase. (a–f) Immune function of NbNTF2A, NbNTF2B, and NbNTF2C. Overexpression of individual *NbNTF2* family members (right half of leaves) and *FGFP* (left half of leaves) via agroinfiltration in *N. benthamiana* detached leaves, followed by *P. parasitica* zoospore inoculation. (a) Trypan blue staining and (b) lesion diameters were analyzed at 48 hpi. (c) qPCR analysis showed relative biomass of *P. parasitica* in both *NbNTF2*s-and *FGFP*-overexpressing leaf halves (n = 6). *NbNTF2* silencing and *GUS* silencing in detached *N. benthamiana* leaves inoculated with *P. parasitica* zoospores. (d) Trypan blue staining and lesion diameters were analyzed at 48 hpi. qPCR analysis showed relative biomass of *P. parasitica* in *NbNTF2*- as compared to *GUS*-silenced leaves. Data were analyzed with Student's *t* test. $*P \le 0.05$, $**P \le 0.01$, ns, no significant difference (n = 6). (e–m) The 40th glutamine residue serves as a susceptibility switch in SIE141 and NbRan1 binding. (e, f) Overexpression of *NbNTF2A*^{E38K}, *NbNTF2A*^{Q40A}, *NbNTF2A*^{Q40E}, or *FGFP* (left half) via agroinfiltration in detached *N. benthamiana* (right half) leaves, followed by *P. parasitica* zoospore inoculation. Lesion diameters and trypan blue staining were analyzed at 48 hpi. (g) qPCR analysis showed relative biomass of *P. parasitica* (n = 6). Data were analyzed with Student's *t* test. $*P \le 0.05$, $**P \le 0.01$, ns, no significant difference. Bars=1 cm. (h–j) Luciferase complementation assay to detect interactions between (h) SIE141 or (i) NbRan1 and NbNTF2A, NbNTF2A^{E38K}, NbNTF2A^{Q40A}, and NbNTF2A^{Q40E}. Nluc-SIE141 or Nluc-NbRan1 and Cluc-NbNTF2A, Cluc-NbNTF2A^{E38K}, Cluc-NbNTF2A^{Q40A}, Cluc-NbNTF2A^{Q40E} were co-expressed via agroinfiltration. (j) Empty vector of Nluc was co-expressed with Cluc-NbNTF2A^{E38K}/Cluc-NbNTF2A^{Q40A}/Cluc-NbNTF2A^{Q40E} while Nluc-NbRan1 and Cluc-NbNTF2A co-expression served as positive control. (k) Bimolecular fluorescence complementation (BiFC) assay to detect interactions between SIE141 or NbRan1 and mutants of NbNTF2A. CVenus-tagged NbNTF2A, NbNTF2A^{E38K}, NbNTF2A^{Q40A}, NbNTF2A^{Q40E} or empty vector were co-expressed with NVenus-tagged SIE141 or NVenus-tagged NbRan1 vector via agroinfiltration in *N. benthamiana* leaves. Confocal microscopy was conducted at 48 hpi. (l-m) Co-IP assay to determine that SIE141 and NbRan1 could not target NbNTF2A mutated in the 40th glutamine residue. GFP-affinity beads were applied for GFP-fused (l) SIE141 or (m) NbRan1, NbNTF2A^{E38K}, NbNTF2A^{Q40A}, NbNTF2A^{Q40E}, and NbNTF2A (positive control) were detected with a MYC-specific antibody. Protein markers indicate size in kDa.

assess their impact on immune function. AlphaFold 3 prediction indicated no change in protein structures for these 2 mutants (Figures S5h and S5i). Luciferase, BiFC, and co-IP assays revealed that both NbNTF2A^{Q40A} and NbNTF2A^{Q40E} mutants lost their interactions with effector SIE141 and NbRan1 (Fig. 4h–4m, Figures S5j–S5k). It indicated that the glutamine residue at position 40 is critical for NbNTF2A interaction with both SIE141 and NbRan1. On the other hand, we further tested the immune function of the mutants NbNTF2A^{Q40A} and NbNTF2A^{Q40E}. Leaves transiently expressing *NbNTF2A*^{Q40A} and *NbNTF2A*^{Q40E} showed increased plant resistance, as evidenced by reduced pathogen biomass, as compared to the *FGFP* control (Figs 4e–4g).

Collectively, these results demonstrate that the conserved E38 residue of NbNTF2A is not required for its interaction with either SIE141 or NbRan1, whereas the paralog-specific Q40 residue critically determined these associations and affected immune outcomes, suggesting that both NbRan1 and SIE141 likely depend on binding to NbNTF2A and subsequent nuclear translocation to exert their respective immune functions.

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NbRan1-mediated immune activation requires its NbNTF2-dependent nuclear accumulation

Ran-GTPase, which predominantly localizes to the nucleus at steady state, accumulates there via the established function of NTF2 (Steggerda et al. 2000). We hypothesized that NTF2 proteins have similar functions toward Ran-GTPase in plants. C-terminal *GFP*-fused *NbRan1* was transiently overexpressed in *GUS*- or *NbNTF2*-silenced *N. benthamiana* leaves and confocal microscopy revealed that *NbNTF2* silencing significantly reduced nuclear NbRan1-GFP signals (Fig. 5a). Subcellular fractionation followed by immunoblotting confirmed that NbRan1 was undetectable in the nuclear fraction from *NbNTF2*-silenced leaves, consistently demonstrating that *NbNTF2*-silencing abolished NbRan1 transfer to the nucleus (Fig. 5b).

To examine whether NbRan1 and its interaction between NbNTF2 are critical for plant immune responses against *P. parasitica*, we transiently expressed *NbRan1-GFP* by agroinfiltration, with *Flag-GFP* (*FGFP*) as a negative control, in *N. benthamiana*, followed by inoculation of detached leaves with *P. parasitica* zoospores. This analysis indicated that *NbRan1* overexpression enhanced resistance to *P. parasitica* as indicated by reduced pathogen biomass as compared to the *FGFP* control (Fig. 5c and d). Conversely, silencing of *NbRan1* increased plant susceptibility to *P. parasitica*, also indicated by an elevated pathogen biomass relative to the *GUS*-silenced plants (Fig. 5e and 5f). Furthermore, transient expression of *NbRan1-GFP* in *NbNTF2*-silenced plants, followed by pathogen challenge on detached leaves showed that *NbRan1* expression conferred increased plant resistance in *GUS*-silenced but not in *NbNTF2*-silenced plants (Fig. 5g and 5h). These results indicated that the immune function of NbRan1 is dependent on NbNTF2 and its mediation of NbRan1 nuclear translocation.

SIE141 influences NbNTF2-NbRan1 interactions without affecting NbRan1-dependent immunity and its nuclear localization

Since both NbNTF2A^{Q40A} and NbNTF2A^{Q40E} cannot interact with SIE141 and NbRan1, SIE141 might affect its binding to Ran1 via this key residue at position 40. To investigate whether SIE141 influences the NbNTF2A-NbRan1 interaction, we performed BiFC assays in *N. benthamiana*. Co-expression of NbNTF2A-CVenus and NbRan1-NVenus yielded YFP signals. When increasing the concentration of co-infiltrated agrobacterial suspensions for *SIE141-mCherry* transformation, YFP fluorescence initially intensified, reaching a maximum at an OD_{600} value of 0.2, and then gradually declined with increasing agrobacterial cells carrying *SIE141-mCherry*. In contrast, co-expression with *mCherry* alone did not show such changes (Figures S6a–S6d). The fluorescence intensity of mCherry confirmed the dose-dependent expression of SIE141 across treatments (Figures S6a–S6d).

These observations were further substantiated by co-IP assays. Transient co-expression of *NbNTF2A-MYC* and *NbRan1-GFP* via agroinfiltration was performed with gradient concentrations of agrobacterial cells carrying *SIE141-Flag* (OD_{600} values of 0, 0.1, 0.2, 0.3, and 0.4). Low to moderate concentrations of *SIE141* agrobacterial cell suspensions (OD_{600} values of 0.1–0.2) enhanced the amount of NbNTF2A co-precipitated with NbRan1-GFP, whereas higher levels (OD_{600} values of 0.3–0.4) reduced it (Fig. 6a). This effect was not observed with Flag-mCherry controls (Fig. 6b).

These findings indicated that SIE141 affects NbNTF2A-NbRanGDP interactions, which may influence subsequent nuclear accumulation of NbRan1. Co-expression of *SIE141-Flag* and *Flag-mCherry*, with *NbRan1-GFP*, respectively, showed that nuclear NbRan1 levels were not changed in the presence of SIE141 (Fig. 6c). Furthermore, quantification of subcellular fractions of the confocal samples showed that the nuclear abundance of NbRan1, as detected by Western blot analyses with anti-GFP antibodies, exhibited no significant difference in samples overexpressing *SIE141-Flag* as compared to those overexpressing *Flag-mCherry* (Fig. 6d). It suggested that SIE141 manipulated NbNTF2A-NbRan1 interactions in a concentration-dependent manner, enhancing complex formation at low to moderate levels while inhibiting it at higher levels.

To further explore immunity effects of SIE141 on NbRan1-mediated plant immunity, we co-expressed *SIE141* and *mCherry* with *NbRan1-GFP*, respectively, followed by inoculation with *P. parasitica* zoospores on detached *N. benthamiana* leaves. Trypan blue staining and biomass quantification revealed similar pathogen development on leaf halves expressing *SIE141* and *NbRan1-GFP* as compared to control halves expressing *mCherry* and *NbRan1-GFP* (Fig. 6e and 6f). These results indicated that *SIE141* overexpression did not affect Ran1-conferred resistance. Furthermore, we tested whether SIE141 could compensate for loss of NbRan1 functions. Overexpression of *SIE141* and *FGFP* in *NbRan1*-silenced plants followed by *P. parasitica* infection showed that *NbRan1*-silencing increased plant susceptibility as tested before (Fig. 6g and 6h). Notably, overexpression of *SIE141* in *NbRan1*-silenced plants significantly reduced pathogen biomass as compared to plants overexpressing *FGFP* and silenced in *NbRan1* while resistance remained similar to *FGFP*-overexpressing, *GUS*-silenced plants (Figs 6g and 6h). Thus, SIE141 partially alleviated enhanced susceptibility caused by *NbRan1* deficiency.

Taken together, these results demonstrated that SIE141 modulated the interaction between NbNTF2A and NbRan1 in a concentration-dependent manner without altering NbRan1 nuclear accumulation. Regarding immunity functions, *SIE141* expression strengthened NbRan1-dependent immune responses against *P. parasitica* and partially compensated for impaired immunity upon *NbRan1* knockdown.

Discussion

Based on our previous study, the endophytic effector SIE141-triggered plant immunity against *Phytophthora* pathogens was found to rely on its nuclear localization (Zhang et al. 2024). However, in contrast to nuclear-localized pathogen effectors, SIE141 does neither contain nuclear localization signals (NLS; eg RxKK) nor target sites for conventional nucleoporins or karyopherins, as revealed by LC-MS/MS analyses (Lim et al. 2024; Zhang et al. 2024). Although most proteins are transported through the nuclear pore complex (NPC) by conventional karyopherin-β mediated mechanisms, Ran systems assisted by NUCLEAR TRANSPORT FACTOR 2 (NTF2) represent alternative nuclear transport pathways that may be employed by nuclear-localized effectors (Cautain et al. 2015).

In this study, we found that SIE141 targets NTF2 family proteins in *Nicotiana benthamiana* and its nuclear localization was impaired by the knock down of *NbNTF2s* (Figs. 1 and 2). Notably, the nuclear import of SIE141 is intrinsically linked to the translocation of its target protein. We previously showed that SIE141 blocks the chloroplast transit peptide of the thioredoxin NbCDSP32, redirecting it to the nucleus to confer disease resistance (Zhang et al. 2024). This study extended this model by demonstrating that the nuclear import of both SIE141 and NbCDSP32 was strictly dependent on NbNTF2 proteins.

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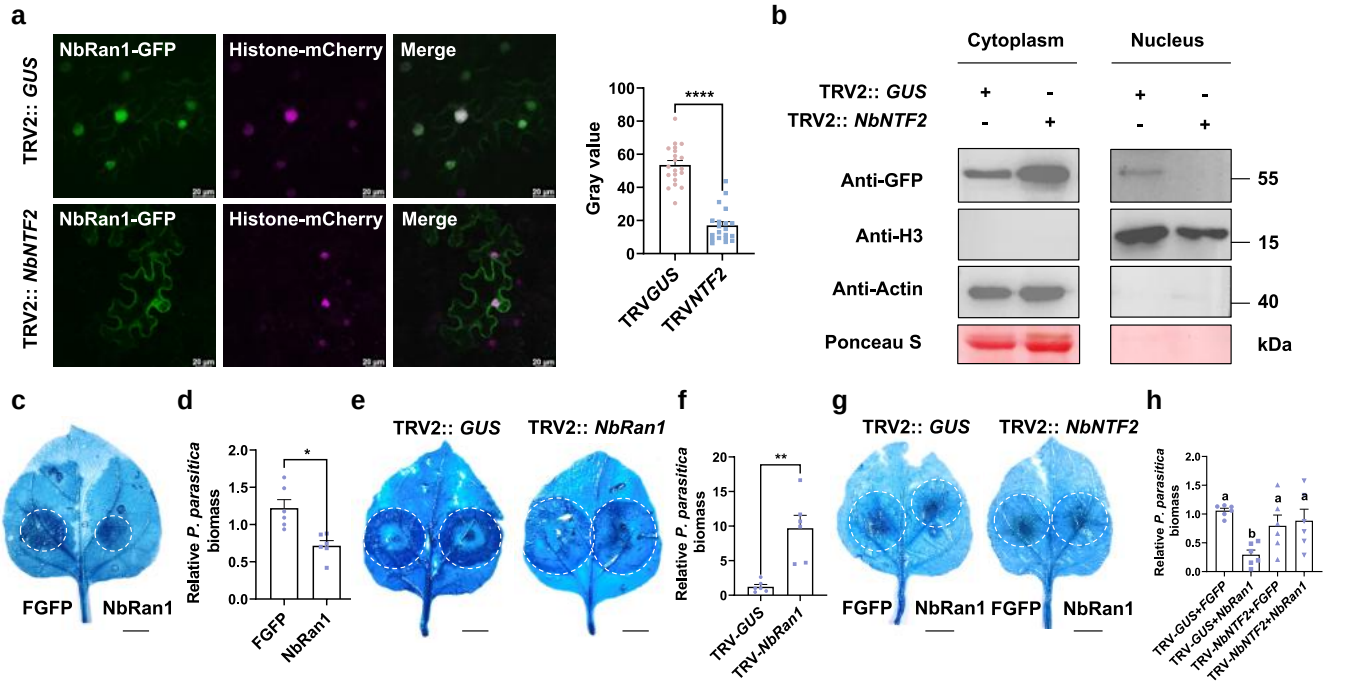


Figure 5 NbRan1-mediated immune activation requires NbNTF2-dependent nuclear accumulation. (a, b) Nuclear accumulation of NbRan1 depends on NbNTF2. (a) NbRan1-GFP was transiently expressed via agroinfiltration in *GUS*-silenced or *NbNTF2*-silenced *N. benthamiana* leaves. Confocal microscopy was conducted at 48 hpi. Bars=20 μm. Fluorescence intensity of NbRan1-GFP signal in individual nuclei (n = 20) of *GUS*- or *NbNTF2*-silenced samples was quantified with imageJ. Data were analyzed with Student's *t* test, ****$P < 0.0001$. (b) Protein levels of NbRan1-GFP were decreased in *NbNTF2*- as compared to *GUS*-silenced samples. Immunoblot analysis for NbRan1-GFP abundance was detected in nuclear extracts from *GUS*- or *NbNTF2*-silenced leaves using a GFP antibody. Histone 3 (H3) antibody was used as internal reference to confirm equal loading of nuclear proteins. Actin antibody and Ponceau S were used as external references. (c–f) NbRan1 positively regulates plant immunity to *P. parasitica*. (c, d) Overexpression of *NbRan1-GFP* via agroinfiltration in *N. benthamiana* detached leaves, followed by *P. parasitica* zoospore inoculation. (c) Trypan blue staining was analyzed at 48 hpi. (d) qPCR data showed relative biomass of *P. parasitica* (n = 6). (e, f) *NbRan1*-silencing in detached *N. benthamiana* leaves inoculated with *P. parasitica* zoospores. (e) Trypan blue staining was analyzed at 48 hpi. (f) qPCR data showed relative biomass of *P. parasitica* (n = 6). Data were analyzed with Student's *t* test. *$P \leq 0.05$, **$P \leq 0.01$. (g, h) NbRan1-mediated immune activation requires NbNTF2. NbRan1-mediated immunity against (g) *P. parasitica* in *N. benthamiana* leaves is lost upon *NbNTF2* silencing indicated by trypan blue staining. (h) Relative pathogen biomasses were quantified at 48 hpi. Data are presented as the mean ± SEM of 3 independent replicates and analyzed with one-way ANOVA (Brown-Forsythe and Welch test, n = 6). Lowercase letters indicate statistical significance between multiple groups at $P < 0.05$. Bars=1 cm.

In *NbNTF2*-silenced plants, neither protein accumulated in nuclei, but instead retained together in chloroplasts (Figs. 2 and 3 and Figure S3). This codependence underscores that SIE141 utilized the NTF2 pathway to co-import an immunity-associated protein complex as a prerequisite for its immunity-conferring effect.

Having established the core trafficking mechanism, we further determined the functional divergence among *NbNTF2* family genes in response to the hemibiotrophic oomycete *P. parasitica*, with NbNTF2A acting as a susceptibility factor against *P. parasitica* (Fig. 4a–4d). Importantly, the paralog-specific Q40 residue of NbNTF2A was essential for immunity against pathogen and critical for its interactions with SIE141 and NbRan1 (Fig. 4e–4m and Figure S5). Although the NTF2 family proteins share high sequence conservation, their functional differences are likely affected by specific cargo of NTF2-Ran transport system. NTF2 proteins primarily function as transport adaptors that maintain the Ran-GTPase gradient essential for nucleocytoplasmic trafficking (Corbett and Silver 1996; Ribbeck et al. 1998). In this context, the cargo proteins that utilize individual NTF2 paralogs together with Ran-GDP/GTP are expected to dictate the ultimate immune output. For instance, nuclear import of key defense regulators such as NPR1 (which requires proper nucleocytoplasmic shuttling for SA-mediated systemic resistance) (Kinkema et al. 2000; Pham et al. 2022) or other virulent effectors from pathogens (Qiang et al. 2023; Lim et al. 2024) could result in different immune phenotypes of NTF2 family proteins.

The primary biological significance of the SIE141-NbNTF2A/B interactions relies on their role as a transport bridge: the NTF2 family proteins are indispensable for nuclear import of SIE141 itself and, more critically, for SIE141-mediated nuclear localization of NbCDSP32. Thus, while *NbNTF2A* overexpression alone promoted plant susceptibility to *P. parasitica*, endogenous NTF2 proteins, including NTF2A, are likely required for supporting nuclear targeting and immune function of the SIE141-CDSP32 complex, leading to enhanced immunity against *P. parasitica*.

An additional possibility is that SIE141 binding to NbNTF2A competitively inhibits nuclear import of pathogen virulence effectors that rely on NTF2A-mediated nuclear trafficking. This competitive mechanism may help explain why SIE141 attenuates plant susceptibility to *P. parasitica*, even though *NbNTF2A* overexpression has a negative effect when tested in isolation (Fig. 4a–4c). Thus, SIE141 does not appear to repress disease resistance in this system; instead, it utilizes host NTF2-dependent transport machinery to redirect CDSP32 complex into the nucleus, where it could promote production of monomeric NPR1 and downstream defense activation (Zhang et al. 2024).

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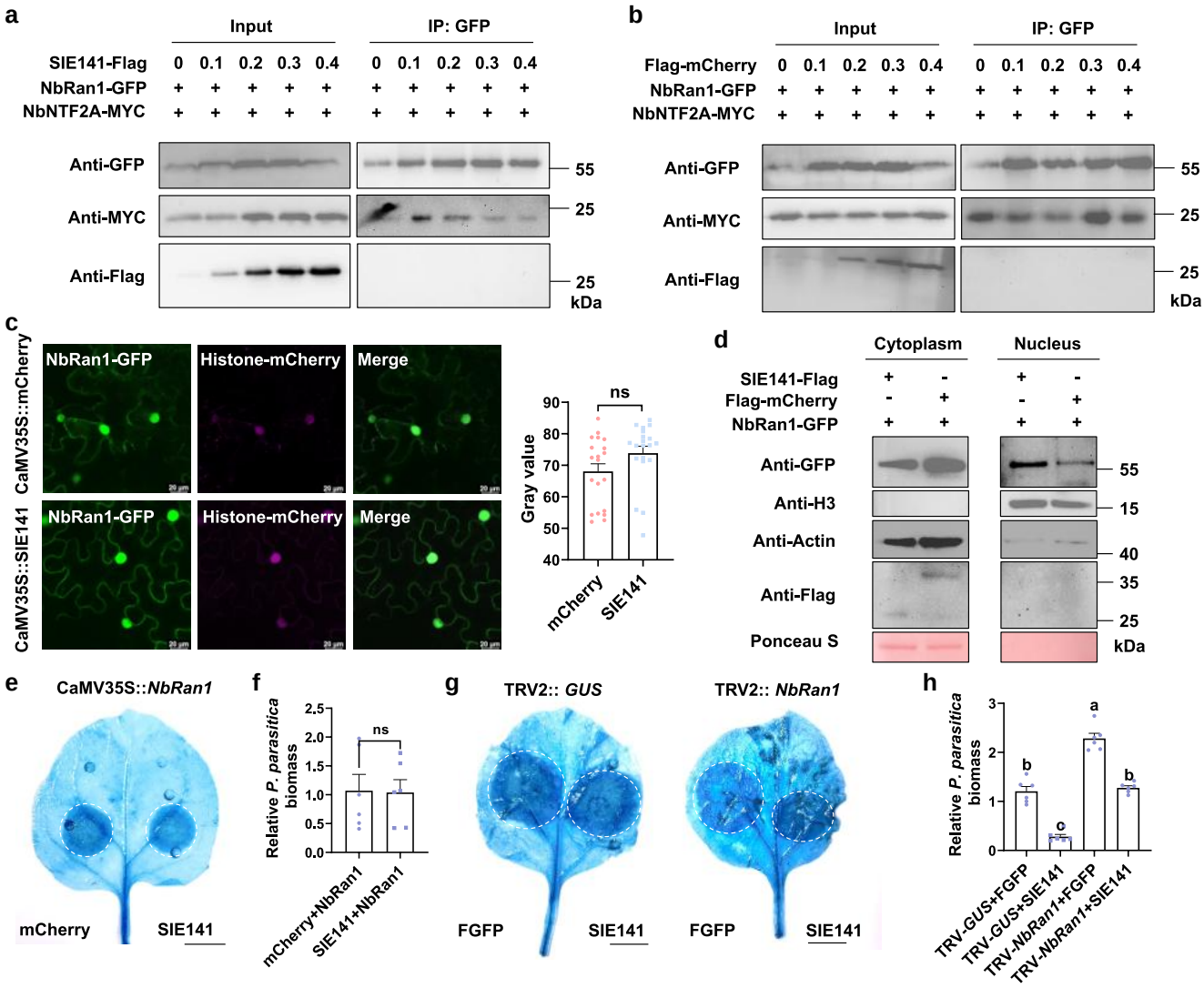


Figure 6 SIE141 influences the NbNTF2-NbRan1 association and modulates NbRan1-dependent immunity without affecting its nuclear localization. (a, b) Co-IP assay indicated a reduction of the NbNTF2A-NbRan1 interaction with increasing (a) SIE141 but not (b) Flag-mCherry (negative control) levels. The protein samples were collected with GFP-Trap for GFP-fused NbRan1, MYC-fused NbNTF2A was detected with a MYC-specific antibody, and Flag-mCherry was detected with a Flag antibody. (c, d) Nuclear accumulation of NbRan1 is not affected by SIE141 overexpression. (c) NbRan1-GFP was co-expressed with mCherry and SIE141 in *N. benthamiana* leaves via agroinfiltration. Histone-mCherry served as nuclear marker. Confocal microscopy was conducted at 48 hpi. Fluorescence intensity of NbRan1-GFP signal in individual nuclei (n = 20) was quantified with ImageJ from samples. Data were analyzed with Student's *t* test. ns, no significant difference. (d) Protein levels of NbRan1-GFP were not affected by co-expression of SIE141-Flag or Flag-mCherry. Immunoblot analysis for NbRan1-GFP abundance was detected using a GFP antibody. Immunoblot analysis for SIE141-Flag or Flag-mCherry abundance were detected using a Flag antibody. Histone 3 (H3) antibody was used as internal reference to confirm equal loading of nuclear proteins. Actin antibody and Ponceau S were used as external references. (e, f) Overexpression of SIE141 or mCherry (control) with NbRan1 in *N. benthamiana* leaves via agroinfiltration, followed by *P. parasitica* zoospore inoculation. (e) Trypan blue staining was conducted at 48 hpi. (f) qPCR data show relative biomass of *P. parasitica.* Data are presented as the mean ± SEM of 6 independent replicates and analyzed with Student's *t* test (paired test, n = 6); ns, no significant difference. (g, h) SIE141 partially compensates for immune deficiency caused by *NbRan1*-knockdown. FGFP and SIE141 were expressed in *GUS* or *NbRan1*-silenced *N. benthamiana* leaves, followed by *P. parasitica* zoospore inoculation. (g) Trypan blue staining was conducted at 48 hpi. (h) qPCR data show relative biomass of *P. parasitica.* Data are presented as the mean ± SEM of 3 independent replicates and analyzed with one-way ANOVA (Brown-Forsythe and Welch test, n = 6). Lowercase letters indicate statistical significance between multiple groups at $P < 0.05$. Bars=1 cm.

In addition, CDSP32 is a chloroplast-localized thioredoxin primarily involved in redox homeostasis regulation (Pant et al. 2020; Sun et al. 2020). Unlike TMX2, a thioredoxin family protein reported to interact with importin-β, Ran, and nuclear pore components to regulate the nucleocytoplasmic Ran gradient and protein import (Oguro and Imaoka 2019), CDSP32 does not interact directly with NbRan1 (Figure S6e). Consistent with this, CDSP32 is not expected to directly participate in the regulation of the Ran concentration gradient. However, Ran-GDP/GTP is involved in nuclear accumulation of NPR1 and functions as a regulation node of SA-induced leaf senescence (Pham et al. 2022). Overexpression of *AtRAN1* positively regulates SA signaling and disease resistance against Pst DC3000 (Xu et al. 2021). Given that SIE141's targeting increased CDSP32 oxidoreductase function in NPR1 depolymerization (Zhang et al. 2024),

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we proposed that the monomerized NPR1 may serve as a molecular hub that integrates the SIE141-CDSP32 redox module with the Ran-dependent nucleocytoplasmic transport and transcriptional machinery in the SA signaling pathway. Further testing of this hypothesis remains an exciting direction for future studies.

Moreover, SIE141 influenced NbNTF2A-NbRan1 binding in a concentration-dependent manner, progressively reducing the interaction at higher doses as demonstrated by co-IP assays (Fig. 6a and 6b). Such dosage sensitivity aligns with NTF2 dysregulation in other systems, where it modulated stress responses, for instance, through soybean GmNTF2B-1 interactions with oxidoreductases to mitigate reactive oxygen species under drought, thereby enhancing drought tolerance (Chen et al. 2021). On the other hand, NbRan1 participation in plant immunity against *P. parasitica* was shown to depend on its nuclear translocation mediated by NbNTF2 (Fig. 5). SIE141 did neither interact with NbRan1, nor affect its nuclear accumulation (Fig. 6a–6d). Further immune function assays determined that *SIE141* overexpression did not disturb NbRan1-dependent immune responses against *P. parasitica* and partially compensated for impaired immunity in the absence of NbRan1 (Fig. 6e–6h). Our analyses thus revealed that SIE141 promoted plant immunity, likely through direct targeting of NbNTF2A to import CDSP32 into the nucleus, rather than affecting plant immune responses dependent on the NTF2–Ran-GTPase system (Fig. 7).

Overall, this refined strategy contrasts with that of pathogenic effectors. Pathogen effectors usually directly target key components of the nuclear trafficking system to suppress plant immunity. For instance, *Magnaporthe oryzae* effector MoHTR1 targeting rice importin α, and *Xanthomonas oryzae* TAL effectors targeting rice importin IMPa1b (Peng et al. 2022; Lim et al. 2024). In this research, we elucidated a previously unknown mechanism in symbiont-induced plant immunity where the endophytic effector SIE141 exploited the conserved host NTF2 pathway to orchestrate its nuclear co-import and its redox target CDSP32 to finally promote plant resistance against *P. parasitica* (Fig. 7). Within this pathway, SIE141 engaged a susceptibility switch (Q40 of NbNTF2A) in a dose-dependent manner to modulate transport dynamics without broadly perturbing host immunity that depends on the NTF2-Ran1 nuclear transportation system. These findings elucidated a mechanism whereby symbiotic effectors harness host nucleocytoplasmic transport to activate immunity, providing new insights into molecular symbioses and targeted engineering of crop resilience against diseases.

Material and methods

Strains, plant materials, and growth conditions

Nicotiana benthamiana were grown at 23 °C under a 13/11 h light/dark cycle. Transient expression assays and VIGS assays were performed with *Agrobacterium tumefaciens* GV3101. *Saccharomyces cerevisiae* strains AH109 were used for yeast transformations and mating. Yeast was cultured at 30 °C on 2× YPDA media. Pathogen inoculation assays used *Phytophthora parasitica* strain Pp042, which was cultured in carrot juice (CA) medium at 23 °C (Pan et al. 2016).

Plasmid construction

SIE141 C-terminal GFP or mCherry tag fused constructs used in this study were as reported in previous work (Zhang et al. 2024). The target genes *NbNTF2* family members were amplified from *N. benthamiana*. Each gene was then fused with a 4×Myc tag and inserted into pART27 at the *Xho*I-*Hind*III sites. The resulting plant expression vectors were used for co-IP or transient expression assays. *NbNTF2A*, *NbNTF2B* and *NbNTF2C* were fused with GFP tag at C-terminus and inserted into pART27 at the *Xho*I-*EcoR*I sites for subcellular localization observation. To generate VIGS constructs, a 200 bp fragment (based on individual gene length) of *NbNTF2s* was amplified from *N. benthamiana* cDNA and cloned into the binary vector pTRV2 at the *BamH*I-*Sal*I sites. All constructs were sequenced by Tsingke (Beijing, China). Primers used for plasmid construction are shown in Table S2.

Pathogen infection assays

Maintenance and zoospore production of *P. parasitica* strain Pp042 was conducted as previously described (Lu et al. 2020). For *N. benthamiana* detached leaves inoculation assay, 20 μL of Pp042 zoospores (10^5 spores/ml) suspension was directly applied to the abaxial side of detached leaves. Leaves were stained with trypan blue at 48 h postinoculation (hpi) to observe lesion area. Infected leaf samples were collected at 48 hpi followed with DNA extraction and qPCR (quantitative real-time PCR) using previously reported methods (Yang et al. 2020).

Transient expression and VIGS assays in *N. benthamiana*

Agrobacterium transient expression assays were carried out in 4- to 6-week-old *N. benthamiana* as previously described (Li et al. 2019). VIGS assays were performed using a previously described method (Du et al. 2021). The *NbNTF2 s* fragment was cloned into TRV2, with a TRV2 vector containing *GUS* serving as a control. The silencing effect was monitored in phytoene desaturase (PDS)-silenced plants as an indicator for the properly silenced leaves. The lower leaves of 6-leaf stage *N. benthamiana* were infiltrated with a mixture of *Agrobacterium* cultures containing TRV1 and TRV2 in equal proportions (final $OD_{600} = 0.25$). VIGS plants were analyzed 3 to 5 weeks after infiltration.

Yeast two-hybrid assay

The Matchmaker Two-Hybrid System 3 protocol (Clontech) was used for all Y2H assays. The effector gene *SIE141* was inserted into the pGBKT7 vector as the bait. The coding sequences corresponding to NbNTF2 family genes were cloned into the pGADT7 vector as prey. The pGBKT7-SIE141 and pGADT7 vectors, each containing a selection gene, were co-transformed into the yeast strain AH109. Transformations were verified by plating on SD/-Trp-Leu media (DDO). Depending on the experiment, interactions were confirmed based on growth on SD/-Trp-Leu-His media (TDO) yield α-galactosidase activity (X-α-gal).

Bimolecular fluorescence complementation assay

BiFC assays were performed as previously described (Li et al. 2019). NVenus was fused to the C-terminal of SIE141 or NbRan1 and CVenus was fused to individual NbNTF2 s and their mutants. Individual *Agrobacterium* cultures were transformed with each vector, then the appropriate pairs of *Agrobacterium* cultures were transiently co-infiltrated into *N. benthamiana* leaves. Fluorescence was observed at 48 hours after infiltration with confocal microscopy.

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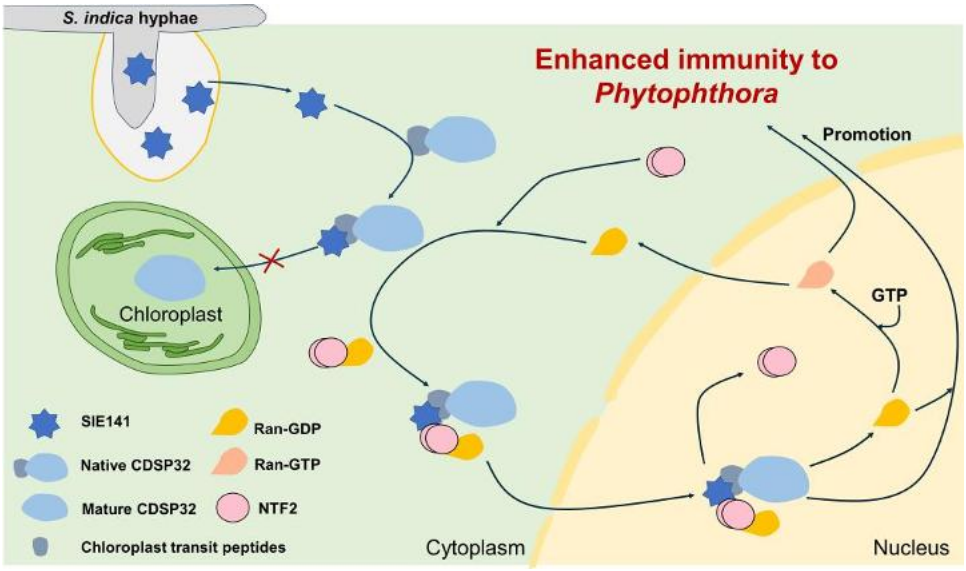


Figure 7 Proposed working model for the nuclear targeting of SIE141 and its target CDSP32 by NTF2 family proteins. SIE141 targets plant NTF2 family proteins to exploit the nucleocytoplasmic transport function of the NTF2-Ran system, enabling nuclear translocation of its target CDSP32 to promote plant immunity against *Phytophthora*.

Co-immunoprecipitation and Western blot assays

Proteins were extracted from *N. benthamiana* leaves expressing co-IP constructs at 2 days postinfiltration. Extractions were conducted with RIPA lysis buffer: 25 mM Tris-HCl, pH 7.5, 150 mM NaCl, 1 mM EDTA, and 0.5% NP-40 with 2% (w/v) polyvinylpolypyrrolidone (PVPP), 1 mM dithiothreitol (DTT), and 10 μL 100X protease inhibitor cocktail (Sellechchem, USA). Total protein samples were incubated at 4 °C for 2 h with 20 μL Anti-GFP Magarose Beads (SMART LIFSCIENCES). For each sample, beads were prewashed 3 times with IP buffer (10 mM Tris-HCl, pH 7.5, 150 mM NaCl, and 0.5 mM EDTA) at 4 °C, 1,200 × *g*, 3 min for 3 times before incubation. The resuspended GFP-Trap beads were mixed with 6× loading buffer and boiled for 10 min at 95 °C to dissociate the immunocomplexes prior to sodium dodecyl sulfate (SDS)-polyacrylamide gel electrophoresis (PAGE). SDS-PAGE was performed using a Fast Preparation Kit (Epizyme Biomedical Technology Co.). Antibodies: anti-MYC (Abmart, #333782), anti-GFP (Abmart, #333077), anti-mCherry (Abmart, M40012), Anti-Luciferase antibody (Sigma, L0159; Sigma, L2164) and goat anti-mouse (Abmart, #M21004) were used for Western blot as instructed by the manufacturer (Abmart, China; ABclonal, China). Detailed Catalog numbers for relevant reagents used are listed in Table S5.

Nuclear and cytoplasm protein extraction

NbNTF2A and *GUS*-silenced *N. benthamiana* leaves expressing *SIE141-GFP* or *N. benthamiana* leaves *SIE141-GFP* together with either *NbNTF2A-mCherry* or *mCherry* were ground with liquid nitrogen. 10 mL precooled extraction buffer I (0.4 M sucrose, 10 mM pH = 8.0 Tris-HCl, 10 mM $MgCl_2$, 5 mM β-ME (2-mercaptoethanol), 0.1 mM PMSF, TrixtonX-100 1% with 1× protease inhibitor cocktail) were added, followed with vortex and spinning for 30 min with rotary mixer. The extracted liquid was filtrated with gauze, centrifugation at 4,000 × *g*, 20 min, 4 °C. 1 mL extraction buffer II (0.25 M sucrose, 10 mM pH = 8.0 Tris-HCl, 10 mM $MgCl_2$, 5 mM β-ME, 0.1 mM PMSF, TrixtonX-100 1% with 1× protease inhibitor cocktail) to suspend the pellet. The sample was centrifuged at 12,000 × *g*, 10 min, 4 °C and washed by 1 mL extraction buffer II for 2 to 3 times. After the last centrifuge, the supernatant was removed, 600 μL extraction buffer III (1.7 M sucrose, 10 mM pH = 8.0 Tris-HCl, 2 mM $MgCl_2$, 5 mM β-ME, 0.1 mM PMSF, TrixtonX-100 0.15% with 1× protease inhibitor cocktail) were added to resuspend the pellet. Carefully, the 600 μL suspension was added to 600 μL new extraction buffer III and centrifuged for 60 min, 14,000 × *g*, 4 °C. The supernatant was removed and resuspended with 700 μL Nuclei Lysis buffer (50 mM pH = 8.0 Tris-HCl, 10 mM EDTA, 1% SDS, 2 mM PMSF, with 1× protease inhibitor cocktail) to get the proteins in nuclear fraction.

For the cytoplasmic fraction, the supernatant from the initial centrifugation was collected and incubated with 1 mL of RIPA lysis buffer. Precooled acetone was then added to the mixture (acetone: extract = 4:1, v/v), and the sample was incubated at −20 °C overnight. The resulting pellet was collected by centrifugation at 12,000 × *g* for 15 min at 4 °C and resuspended in 1 mL of 10 mM HEPES buffer. Protein loading buffer was subsequently added to the cytoplasmic and nuclear fractions from each sample prior to SDS-PAGE analysis. Detailed Catalog numbers for relevant reagents used are listed in Table S5.

Data analyses

All statistical analyses were performed using GraphPad Prism version 8.0 (GraphPad Software, USA; RRID: SCR_002798). For paired data obtained from left and right leaf infiltrations/infections within the same plant, a paired Student's *t* test was applied (Figs 4b–4c, 4h–4e, 5d–5f, 6f, etc.) and confocal microscopy fluorescence intensity measurements (Figs 5a and 6c).

For comparisons among multiple independent groups, one-way ANOVA with Brown-Forsythe and Welch tests was used, followed by post hoc multiple comparisons. Different letters indicate significant differences at $P < 0.05$. These analyses were applied in the bacterial

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infection experiments (Figs 5h and 6h) and confocal microscopy intensity quantification (Fig. 2b).

Data are presented as mean ± standard error of the mean (SEM), as indicated in the figure legends. All experiments were repeated at least 3 times with similar results unless otherwise stated.

For IP-MS data, detailed information of the NbNTF2 family proteins reported here (accession numbers, number of unique peptides, sequence coverage, experimental q-values, and other MS parameters) is provided in Table S1. The full IP-MS dataset (including raw files) is available from the corresponding authors upon reasonable request.

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Author contributions

W.S. and P.S. conceived and supervised the research. Y.Z. designed the experiments and wrote the manuscript, with contribution from all authors. Y.Z. and Z.L. conducted most experiments. Y.Z. performed yeast two-hybrid and luciferase assays. X.M. and Y.L. contributed to vector construction. Y.Y. contributed to schematic model drawing. Z.Y., Y.Y., and Y.M. contributed to data analyses. P.S., L.R., and Y.M. helped with figure arrangements and text proofreading.

Supplementary material

Supplementary material is available at *Plant Physiology* online.

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Conflicts of interests

The authors declare no competing interests.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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